

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     for (int i = 0; i < n; i++) {
13         for (int j = i + 1; j < n; j++) {
14             if (a[i] == a[j]) {
15                 printf("%d", a[i]);
16                 return 0;
17             }
18         }
19     }
20
21     return 0;
22 }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

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Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

```
1
3 10 17 57
6 2 7 10 15 57 246
```

Output:

```
10 57
```

Input:

```
1
6 1 2 3 4 5 6
2 1 6
```

Output:

```
1 6
```

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int T, n, m, i, j;
5     scanf("%d", &T);
6
7     while (T--) {
8         scanf("%d", &n);
9         int a[n];
10        for(i=0;i<n;i++) scanf("%d",&a[i]);
11
12        scanf("%d", &m);
13        int b[m];
14        for(i=0;i<m;i++) scanf("%d",&b[i]);
15    }
```

```
15 |
16 |     i=j=0;
17 |     while(i<n && j<m) {
18 |         if(a[i]==b[j]) {
19 |             printf("%d ",a[i]);
20 |             i++; j++;
21 |         }
22 |         else if(a[i]<b[j]) i++;
23 |         else j++;
24 |     }
25 |     printf("\n");
26 | }
27 | }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

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Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1
2  #include <stdio.h>
3
4  int main() {
5      int T;
6      scanf("%d", &T);
7
8      while (T-- > 0) {
9          int N1, N2;
10         scanf("%d", &N1);
11
12         int a[N1];
13         for (int i = 0; i < N1; i++) {
14             scanf("%d", &a[i]);
15         }
16     }
17 }
```

```

15     }
16
17     scanf("%d", &N2);
18
19     int b[N2];
20     for (int i = 0; i < N2; i++) {
21         scanf("%d", &b[i]);
22     }
23
24     int i = 0, j = 0, first = 1;
25
26     while (i < N1 && j < N2) {
27         if (a[i] == b[j]) {
28             if (!first) {
29                 printf(" ");
30             }
31             printf("%d", a[i]);
32             first = 0;
33             i++;
34             j++;
35         } else if (a[i] < b[j]) {
36             i++;
37         } else {
38             j++;
39         }
40     }
41
42     printf("\n");
43 }
44
45 return 0;
46 }
47
48
49

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

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Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5
6      scanf("%d", &n);
7
8      int a[n];
9
10     for (int i = 0; i < n; i++) {
11         scanf("%d", &a[i]);
12     }
13
14     scanf("%d", &k);
15
16     int i = 0, j = 1;
17
18     while (i < n && j < n) {
19         if (i == j) {
20             j++;
21         }
22         else if (a[j] - a[i] == k) {
23             printf("1");
24             return 0;
25         }
26         else if (a[j] - a[i] < k) {
27             j++;
28         }
29         else {
30             i++;
31         }
32     }

```

```
33 |
34 |     printf("0");
35 |
36 |     return 0;
37 | }
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

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19             j++;
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26             j++;
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28         else {
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✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

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